

Note to other teachers and users of these slides: We would be delighted if you found this our material useful in giving your own lectures. Feel free to use these slides verbatim, or to modify them to fit your own needs. If you make use of a significant portion of these slides in your own lecture, please include this message, or a link to our web site: <http://www.mmds.org>

Recommender Systems: Content-based Systems & Collaborative Filtering

Mining of Massive Datasets

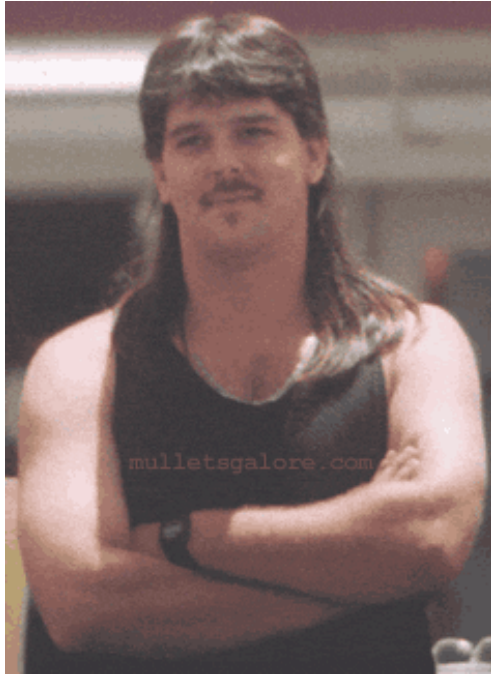
Jure Leskovec, Anand Rajaraman, Jeff Ullman

Stanford University

<http://www.mmds.org>



Example: Recommender Systems



■ Customer X

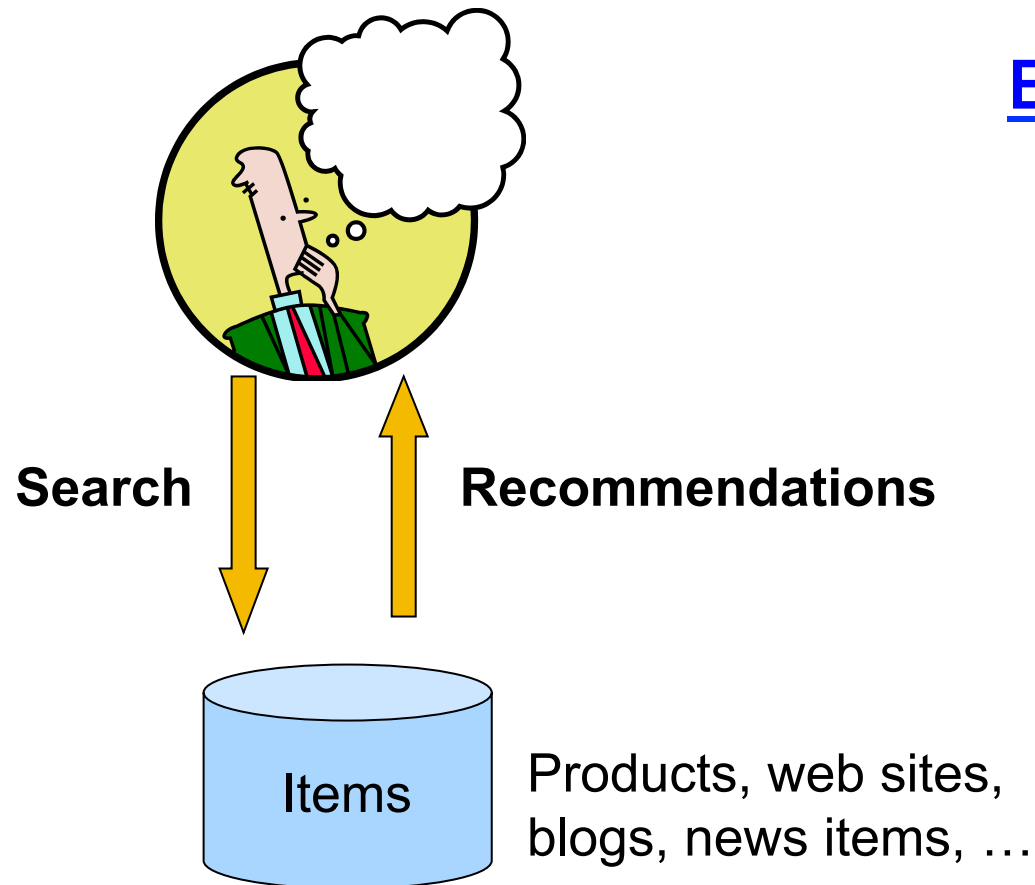
- Buys Metallica CD
- Buys Megadeth CD



■ Customer Y

- Does search on Metallica
- Recommender system suggests Megadeth from data collected about customer X

Recommendations



Examples:

amazon.com.



StumbleUpon



del.icio.us



m o v i e l e n s
helping you find the *right* movies

last.fm™
the social music revolution

Google™
News

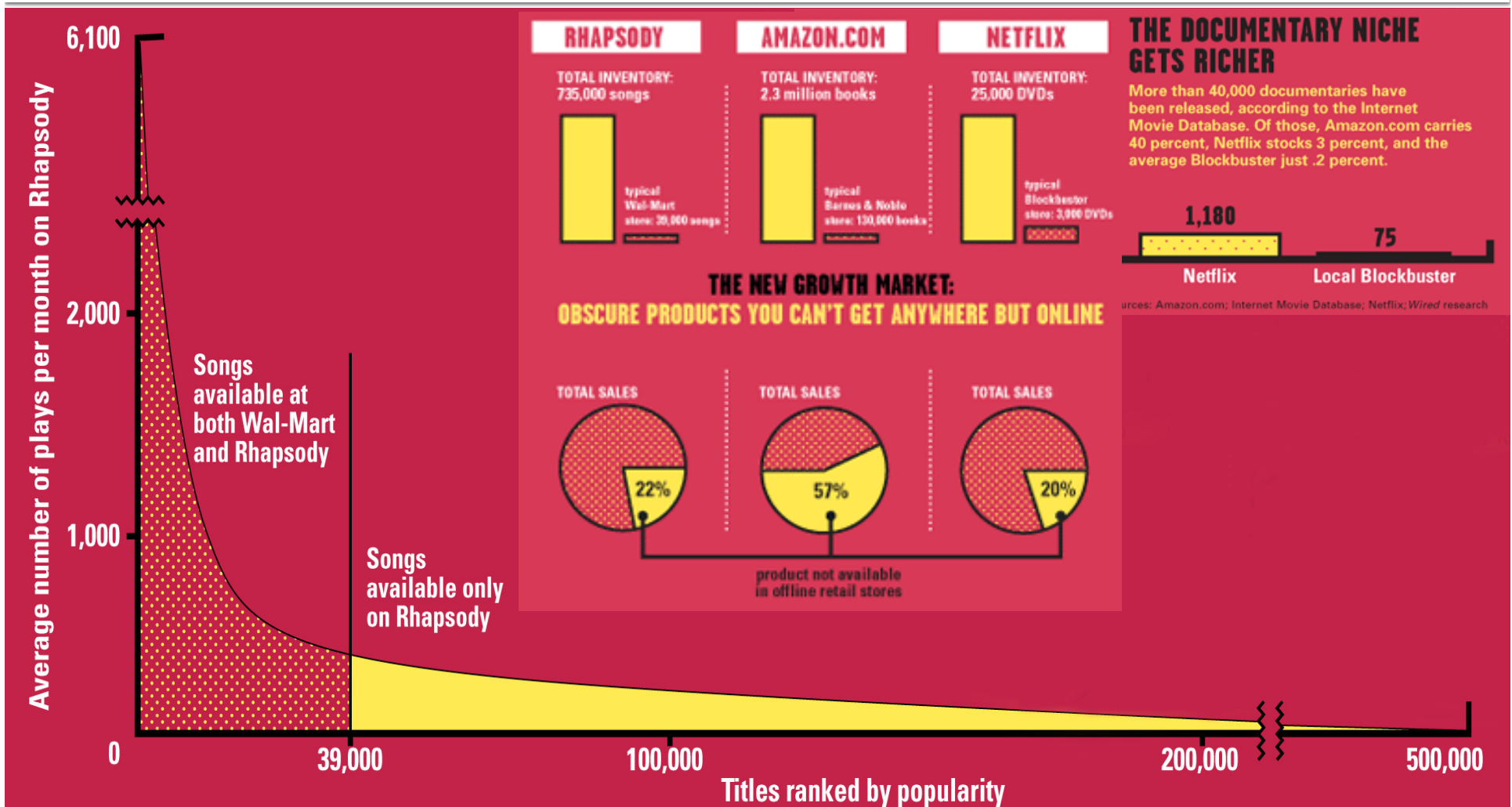
You Tube

XBOX
LIVE

From Scarcity to Abundance

- **Shelf space is a scarce commodity for traditional retailers**
 - Also: TV networks, movie theaters,...
- **Web enables near-zero-cost dissemination of information about products**
 - From scarcity to abundance
- **More choice necessitates better filters**
 - Recommendation engines
 - How **Into Thin Air** made **Touching the Void** a bestseller: <http://www.wired.com/wired/archive/12.10/tail.html>

Sidenote: The Long Tail



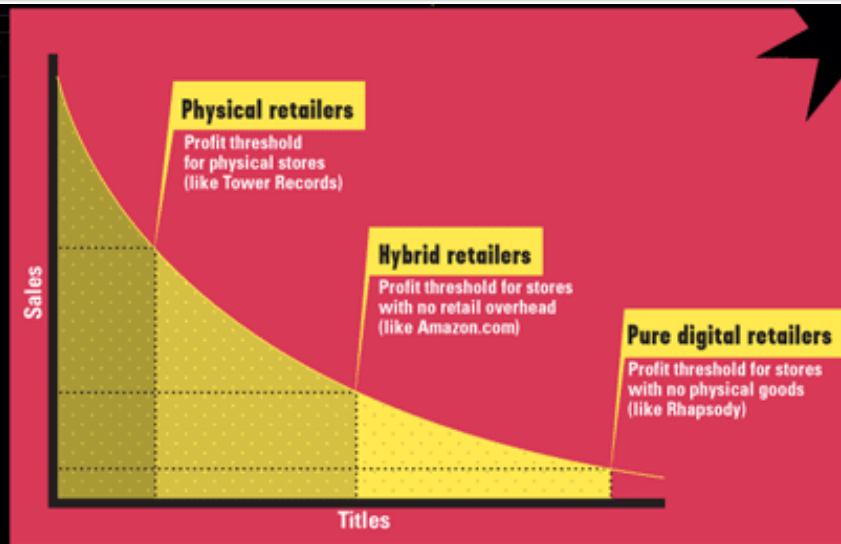
Sources: Erik Brynjolfsson and Jeffrey Hu, MIT, and Michael Smith, Carnegie Mellon; Barnes & Noble; Netflix; RealNetworks
Source: Chris Anderson (2004)

Physical vs. Online

THE BIT PLAYER ADVANTAGE

Beyond bricks and mortar there are two main retail models – one that gets halfway down the Long Tail and another that goes all the way. The first is the familiar hybrid model of Amazon and Netflix, companies that sell physical goods online. Digital catalogs allow them to offer unlimited selection along with search, reviews, and recommendations, while the cost savings of massive warehouses and no walk-in customers greatly expands the number of products they can sell profitably.

Pushing this even further are pure digital services, such as iTunes, which offer the additional savings of delivering their digital goods online at virtually no marginal cost. Since an extra database entry and a few megabytes of storage on a server cost effectively nothing, these retailers have no economic reason not to carry *everything* available.



Read <http://www.wired.com/wired/archive/12.10/tail.html> to learn more!

Types of Recommendations

- **Editorial and hand curated**
 - List of favorites
 - Lists of “essential” items
- **Simple aggregates**
 - Top 10, Most Popular, Recent Uploads
- **Tailored to individual users**
 - Amazon, Netflix, ...

Formal Model

- X = set of **Customers**
- S = set of **Items**
- **Utility function** $u: X \times S \rightarrow R$
 - R = set of ratings
 - R is a totally ordered set
 - e.g., **0-5** stars, real number in **[0,1]**

Utility Matrix

	Avatar	LOTR	Matrix	Pirates
Alice	1		0.2	
Bob		0.5		0.3
Carol	0.2		1	
David				0.4

Key Problems

- **(1) Gathering “known” ratings for matrix**
 - How to collect the data in the utility matrix
- **(2) Extrapolate unknown ratings from the known ones**
 - Mainly interested in high unknown ratings
 - We are not interested in knowing what you don't like but what you like
- **(3) Evaluating extrapolation methods**
 - How to measure success/performance of recommendation methods

(1) Gathering Ratings

■ Explicit

- Ask people to rate items
- Doesn't work well in practice – people can't be bothered

■ Implicit

- Learn ratings from user actions
 - E.g., purchase implies high rating
- What about low ratings?

(2) Extrapolating Utilities

- **Key problem:** Utility matrix U is **sparse**
 - Most people have not rated most items
 - **Cold start:**
 - New items have no ratings
 - New users have no history
- **Three approaches to recommender systems:**
 - 1) Content-based
 - 2) Collaborative
 - 3) Latent factor based

} **Today!**

Content-based Recommender Systems

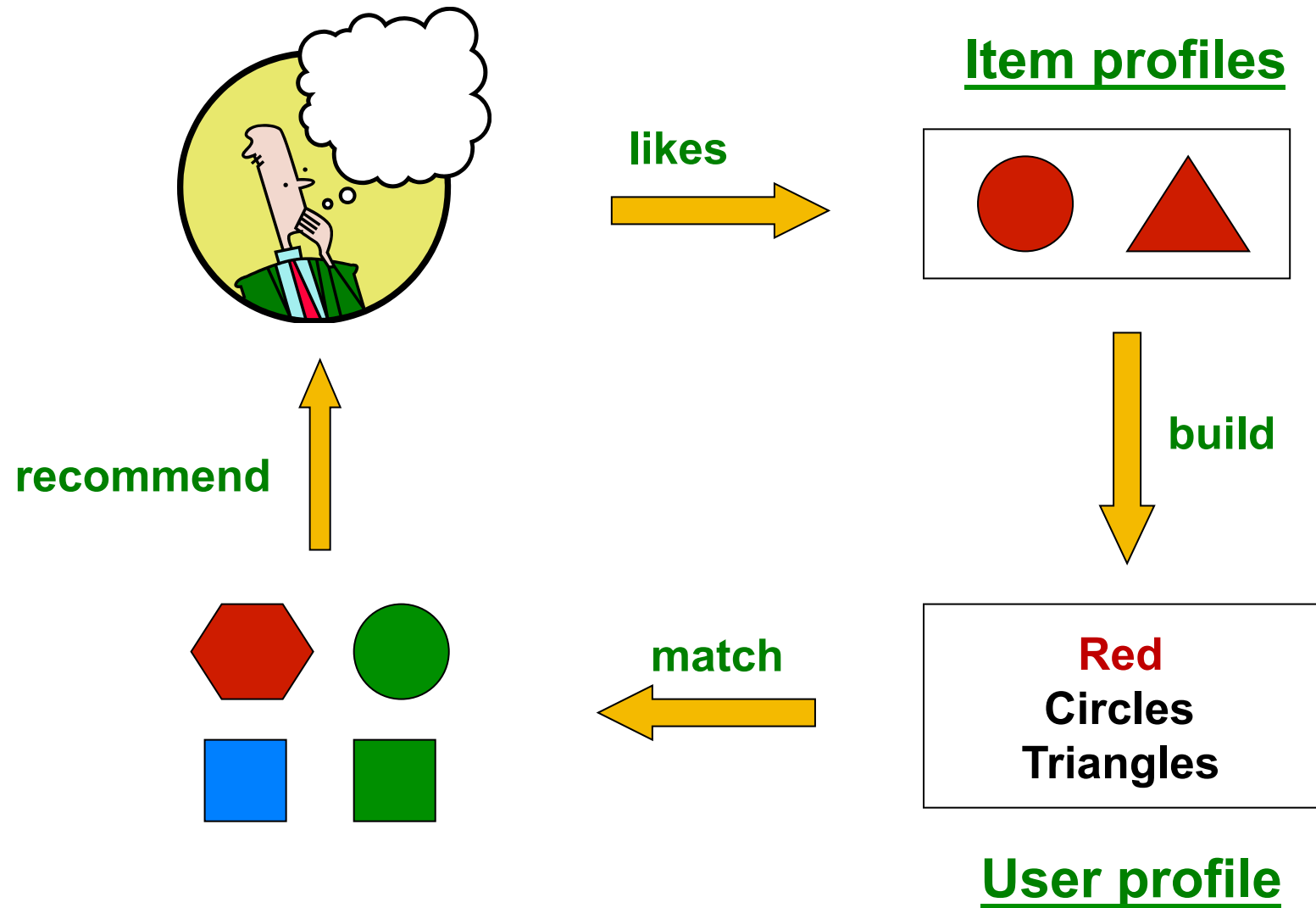
Content-based Recommendations

- **Main idea:** Recommend items to customer x similar to previous items rated highly by x

Example:

- **Movie recommendations**
 - Recommend movies with same actor(s), director, genre, ...
- **Websites, blogs, news**
 - Recommend other sites with “similar” content

Plan of Action



Item Profiles

- For each item, create an **item profile**
- **Profile is a set (vector) of features**
 - **Movies:** author, title, actor, director,...
 - **Text:** Set of “important” words in document
- **How to pick important features?**
 - Usual heuristic from text mining is **TF-IDF**
(Term frequency * Inverse Doc Frequency)
 - **Term ... Feature**
 - **Document ... Item**

Sidenote: TF-IDF

f_{ij} = frequency of term (feature) i in doc (item) j

$$TF_{ij} = \frac{f_{ij}}{\max_k f_{kj}}$$

Note: we normalize TF to discount for “longer” documents

n_i = number of docs that mention term i

N = total number of docs

$$IDF_i = \log \frac{N}{n_i}$$

TF-IDF score: $w_{ij} = TF_{ij} \times IDF_i$

Doc profile = set of words with highest **TF-IDF** scores, together with their scores

User Profiles and Prediction

- **User profile possibilities:**

- Weighted average of rated item profiles
- **Variation:** weight by difference from average rating for item
- ...

- **Prediction heuristic:**

- Given user profile $\mathbf{x}$ and item profile $\mathbf{i}$, estimate $u(\mathbf{x}, \mathbf{i}) = \cos(\mathbf{x}, \mathbf{i}) = \mathbf{x} \cdot \mathbf{i} / \|\mathbf{x}\| \cdot \|\mathbf{i}\|$

Pros: Content-based Approach

- **+: No need for data on other users**
 - No cold-start or sparsity problems
- **+: Able to recommend to users with unique tastes**
- **+: Able to recommend new & unpopular items**
 - No first-rater problem
- **+: Able to provide explanations**
 - Can provide explanations of recommended items by listing content-features that caused an item to be recommended

Cons: Content-based Approach

- —: Finding the appropriate features is hard
 - E.g., images, movies, music
- —: Recommendations for new users
 - How to build a user profile?
- —: Overspecialization
 - Never recommends items outside user's content profile
 - People might have multiple interests
 - Unable to exploit quality judgments of other users

Recommender Systems

Collaborative Filtering

Mining of Massive Datasets

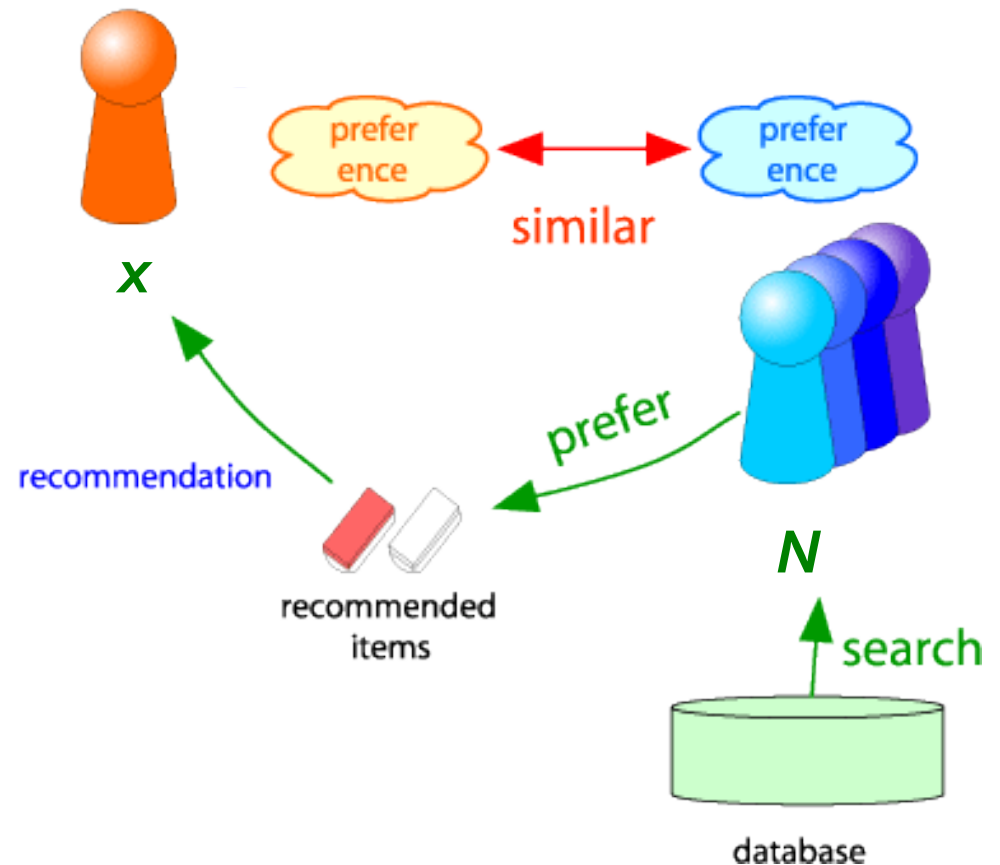
Leskovec, Rajaraman, and Ullman

Stanford University



Collaborative Filtering

- Consider user x
- Find set N of other users whose ratings are “similar” to x ’s ratings
- Estimate x ’s ratings based on ratings of users in N



Similar Users (1)

	HP1	HP2	HP3	TW	SW1	SW2	SW3
<i>A</i>	4			5	1		
<i>B</i>	5	5	4				
<i>C</i>				2	4	5	
<i>D</i>		3					3

- Consider users $\mathbf{x}$ and $\mathbf{y}$ with rating vectors $\mathbf{r}_x$ and $\mathbf{r}_y$
- We need a similarity metric $\text{sim}(\mathbf{x}, \mathbf{y})$
- Capture intuition that $\text{sim}(A, B) > \text{sim}(A, C)$

Option 1: Jaccard Similarity

	HP1	HP2	HP3	TW	SW1	SW2	SW3
<i>A</i>	4			5	1		
<i>B</i>	5	5	4				
<i>C</i>				2	4	5	
<i>D</i>		3					3

- $\text{sim}(A,B) = |r_A \cap r_B| / |r_A \cup r_B|$
- $\text{sim}(A,B) = 1/5$; $\text{sim}(A,C) = 2/4$
 - $\text{sim}(A,B) < \text{sim}(A,C)$
- Problem: Ignores rating values!

Option 2: Cosine similarity

	HP1	HP2	HP3	TW	SW1	SW2	SW3
<i>A</i>	4			5	1		
<i>B</i>	5	5	4				
<i>C</i>				2	4	5	
<i>D</i>		3					3

- $\text{sim}(A,B) = \cos(r_A, r_B)$
- $\text{sim}(A,B) = 0.38$, $\text{sim}(A,C) = 0.32$
 - $\text{sim}(A,B) < \text{sim}(A,C)$, but not by much
- Problem: treats missing ratings as negative

Option 3: Centered cosine

- Normalize ratings by subtracting row mean

	HP1	HP2	HP3	TW	SW1	SW2	SW3
<i>A</i>	4			5	1		
<i>B</i>	5	5	4				
<i>C</i>				2	4	5	
<i>D</i>		3					3

	HP1	HP2	HP3	TW	SW1	SW2	SW3
<i>A</i>	$2/3$			$5/3$	$-7/3$		
<i>B</i>	$1/3$	$1/3$	$-2/3$				
<i>C</i>				$-5/3$	$1/3$	$4/3$	
<i>D</i>		0					0

Centered Cosine similarity (2)

	HP1	HP2	HP3	TW	SW1	SW2	SW3
A	2/3			5/3	-7/3		
B	1/3	1/3	-2/3				
C				-5/3	1/3	4/3	
D		0					0

- $\text{sim}(A,B) = \cos(r_A, r_B) = 0.09$; $\text{sim}(A,C) = -0.56$
 - $\text{sim}(A,B) > \text{sim}(A,C)$
- Captures intuition better
 - Missing ratings treated as “average”
 - Handles “tough raters” and “easy raters”
- Also known as **Pearson Correlation**

Rating Predictions

- Let r_x be the vector of user x 's ratings
- Let N be the set of k users most similar to x who have also rated item i
- Prediction for user x and item i
- Option 1: $r_{xi} = 1/k \sum_{y \in N} r_{yi}$
- Option 2: $r_{xi} = \sum_{y \in N} s_{xy} r_{yi} / \sum_{y \in N} s_{xy}$
where $s_{xy} = \text{sim}(x,y)$

Item-Item Collaborative Filtering

- So far: **User-user collaborative filtering**
- **Another view: Item-item**
 - For item i , find other similar items
 - Estimate rating for item i based on ratings for similar items
 - Can use same similarity metrics and prediction functions as in user-user model

$$r_{xi} = \frac{\sum_{j \in N(i;x)} s_{ij} \cdot r_{xj}}{\sum_{j \in N(i;x)} s_{ij}}$$

s_{ij} ... similarity of items i and j

r_{xj} ... rating of user x on item j

$N(i;x)$... set items rated by x similar to i

Item-Item CF ($|N|=2$)

users

movies

	1	2	3	4	5	6	7	8	9	10	11	12
1	1		3			5			5		4	
2			5	4			4			2	1	3
3	2	4		1	2		3		4	3	5	
4		2	4		5			4			2	
5			4	3	4	2					2	5
6	1		3		3			2			4	

- unknown rating

- rating between 1 to 5

Item-Item CF ($|N|=2$)

		users											
		1	2	3	4	5	6	7	8	9	10	11	12
movies	1	1		3		?	5			5		4	
	2			5	4			4			2	1	3
	3	2	4		1	2		3		4	3	5	
	4		2	4		5			4			2	
	5			4	3	4	2					2	5
	6	1		3		3			2			4	



- estimate rating of movie 1 by user 5

Item-Item CF ($|N|=2$)

		users													
		1	2	3	4	5	6	7	8	9	10	11	12		
movies	1	1		3		?	5			5		4		$\text{sim}(1,m)$	
	2			5	4			4			2	1	3	1.00	
	<u>3</u>	2	4		1	2		3		4	3	5		<u>0.41</u>	
	4		2	4		5			4			2		-0.18	
	5			4	3	4	2					2	5	-0.10	
	<u>6</u>	1		3		3			2			4		<u>0.59</u>	

Neighbor selection:
Identify movies similar to
movie 1, **rated by user 5**

Here we use Pearson correlation as similarity:
1) Subtract mean rating m_i from each movie i
 $m_1 = (1+3+5+5+4)/5 = 3.6$
 row 1: [-2.6, 0, -0.6, 0, 0, 1.4, 0, 0, 1.4, 0, 0.4, 0]
 2) Compute cosine similarities between rows

Item-Item CF ($|N|=2$)

		users													
		1	2	3	4	5	6	7	8	9	10	11	12		
movies	1	1		3		?	5			5		4		$\text{sim}(1,m)$	
	2			5	4			4			2	1	3	1.00	
	<u>3</u>	2	4		1	2		3		4	3	5		<u>0.41</u>	
	4		2	4		5			4			2		-0.18	
	5			4	3	4	2					2	5	-0.10	
	<u>6</u>	1		3		3			2			4		-0.31	
														<u>0.59</u>	

Compute similarity weights:

$s_{13}=0.41$, $s_{16}=0.59$

Item-Item CF ($|N|=2$)

		users											
		1	2	3	4	5	6	7	8	9	10	11	12
movies	1	1		3		2.6	5			5		4	
	2			5	4			4			2	1	3
	<u>3</u>	2	4		1	2		3		4	3	5	
	4		2	4		5			4			2	
	5			4	3	4	2					2	5
	<u>6</u>	1		3		3			2			4	

Predict by taking weighted average:

$$r_{15} = (0.41 \cdot 2 + 0.59 \cdot 3) / (0.41 + 0.59) = 2.6$$

Item-Item v. User-User

- In theory, user-user and item-item are dual approaches
- In practice, item-item outperforms user-user in many use cases
- Items are “simpler” than users
 - Items belong to a small set of “genres”, users have varied tastes
 - Item Similarity is more meaningful than User Similarity

Recommender Systems

Implementing Collaborative Filtering

Mining of Massive Datasets

Leskovec, Rajaraman, and Ullman

Stanford University



Pros/Cons of Collaborative Filtering

- **+ Works for any kind of item**
 - No feature selection needed
- **- Cold Start:**
 - Need enough users in the system to find a match
- **- Sparsity:**
 - The user/ratings matrix is sparse
 - Hard to find users that have rated the same items
- **- First rater:**
 - Cannot recommend an unrated item
 - New items, Esoteric items
- **- Popularity bias:**
 - Tends to recommend popular items

Hybrid Methods

- **Add content-based methods to collaborative filtering**
 - Item profiles for new item problem
 - Demographics to deal with new user problem
- **Implement two or more different recommenders and combine predictions**
 - Perhaps using a linear model
 - Example: global baseline + collaborative filtering

Global Baseline Estimate

- Estimate Joe's rating for the movie *The Sixth Sense*
 - Problem: Joe has not rated any movie “similar” to *The Sixth Sense*
- Global Baseline approach
 - Mean movie rating: **3.7 stars**
 - *The Sixth Sense* is **0.5** stars above avg.
 - Joe rates **0.2** stars below avg.
 - **Baseline estimate: $3.7 + 0.5 - 0.2 = 4$ stars**

Combining Global Baseline with CF

- Global Baseline estimate
 - Joe will give *The Sixth Sense* 4 stars
- Local neighborhood (CF/NN)
 - Joe didn't like related movie *Signs*
 - Rated it 1 star below his average rating
- Final estimate
 - Joe will rate *The Sixth Sense* $4 - 1 = 3.5$ stars

CF: Common Practice

Before:

$$r_{xi} = \frac{\sum_{j \in N(i;x)} s_{ij} r_{xj}}{\sum_{j \in N(i;x)} s_{ij}}$$

- Define **similarity** s_{ij} of items i and j
- Select k nearest neighbors $N(i; x)$
 - Items most similar to i , that were rated by x
- Estimate rating r_{xi} as the weighted average:

$$r_{xi} = b_{xi} + \frac{\sum_{j \in N(i;x)} s_{ij} \cdot (r_{xj} - b_{xj})}{\sum_{j \in N(i;x)} s_{ij}}$$

baseline estimate for r_{xi}

$$b_{xi} = \mu + b_x + b_i$$

- μ = overall mean movie rating
- b_x = rating deviation of user x
= (avg. rating of user x) - μ
- b_i = rating deviation of movie i

Item-Item vs. User-User

	Avatar	LOTR	Matrix	Pirates
Alice	1		0.8	
Bob		0.5		0.3
Carol	0.9		1	0.8
David			1	0.4

- In practice, it has been observed that item-item often works better than user-user
- **Why?** Items are simpler, users have multiple tastes

Remarks & Practical Tips

- Evaluation
- Error metrics
- Complexity / Speed

Recommender Systems

Evaluating recommender systems

Mining of Massive Datasets

Leskovec, Rajaraman, and Ullman

Stanford University



Evaluation

movies

users

1	3	4			
	3	5			5
		4	5		5
		3			
		3			
2			2		2
				5	
	2	1			1
	3			3	
1					

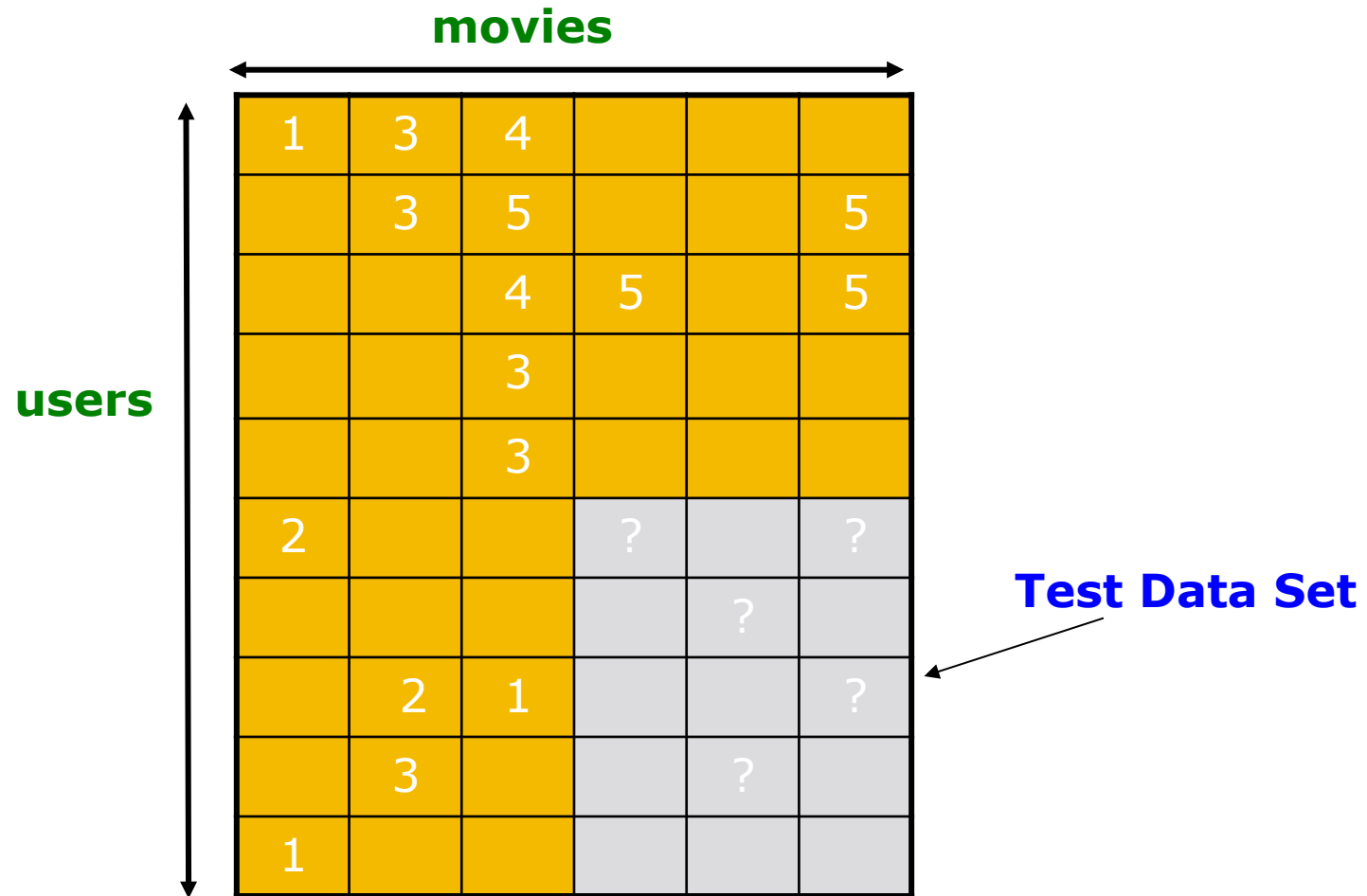
Evaluation

movies

users

1	3	4			
	3	5			5
		4	5		5
		3			
		3			
2			?		?
				?	
	2	1			?
	3			?	
1					

Test Data Set



Evaluating Predictions

- Compare predictions against withheld known ratings (test set T)
- Root-mean-square error (RMSE)

$$\sqrt{\frac{\sum_{(x,i) \in T} (r_{xi} - r_{xi}^*)^2}{N}}$$

where $N = |T|$

r_{xi} is the predicted rating

r_{xi}^* is the actual rating

Problems with Error Measures

- **Narrow focus on accuracy sometimes misses the point**
 - Prediction Diversity
 - Prediction Context
 - Order of predictions
- **In practice, we care only to predict high ratings**
 - RMSE might penalize a method that does well for high ratings and badly for others
 - Alternative: precision at top k
 - Percentage of predictions in the user's top k withheld ratings

Collaborative Filtering: Complexity

- Expensive step is finding k most similar customers: $O(|X|)$
- **Too expensive to do at runtime**
 - Could pre-compute
- Naïve pre-computation takes time $O(k \cdot |X|)$
 - X ... set of customers
- **We already know how to do this!**
 - Near-neighbor search in high dimensions (**LSH**)
 - Clustering
 - Dimensionality reduction

Tip: Add Data

- **Leverage all the data**

- Don't try to reduce data size in an effort to make fancy algorithms work
- Simple methods on large data do best

- **Add more data**

- e.g., add IMDB data on genres

- **More data beats better algorithms**

<http://anand.typepad.com/datawocky/2008/03/more-data-usual.html>